A proof of the Chiu–Ng conjecture on Shabat–Blaschke products

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November 2026

1 The statement

Notation. We write $\mathbb{D}$ for the open unit disc and $\mathbb{H} = \{\tau \in \mathbb{C} : \operatorname{Im}\tau > 0\}$ for the upper half-plane. A *tree* is a plane tree whose vertices are coloured black and white, and in which every edge joins a black vertex to a white one. Two trees are considered the same if an orientation-preserving homeomorphism of the plane carries one onto the other, with the colours. As in [1, Section 7], for $\tau \in \mathbb{H}$ and $q = e^{2\pi i\tau}$ we put

$$\vartheta_2(\tau) = \sum_{m\in\mathbb{Z}} q^{(m+1/2)^2}, \qquad \vartheta_3(\tau) = \sum_{m\in\mathbb{Z}} q^{m^2}, \qquad \vartheta_0(\tau) = \sum_{m\in\mathbb{Z}} (-1)^m q^{m^2},$$

and

$$\sqrt{k(\tau)} = \frac{\vartheta_2(\tau)}{\vartheta_3(\tau)}, \qquad \omega_1(\tau) = \vartheta_3(\tau)^2, \qquad s_n(\tau) = n\tau.$$

The products B_τ. Let T be a tree with n edges. We call $m_1, \dots, m_r$ the valencies of its black vertices and $\mu_1, \dots, \mu_s$ those of its white vertices. Then $\sum m_i = \sum \mu_j = n$, and $r + s = n + 1$ because a tree with n edges has $n+1$ vertices. We assume that T is not a star, so that it has a black vertex and a white vertex of valency at least 2. For τ on the positive imaginary axis we put

$$c = c(\tau) = \sqrt{k(n\tau)} = \frac{\vartheta_2(n\tau)}{\vartheta_3(n\tau)},$$

which is a real number between 0 and 1 (Lemma 2). Kenneth Chung Tak Chiu and Tuen Wai Ng [1] consider the Blaschke products B of degree n with the following two properties.

- The critical values of B in $\mathbb{D}$ are exactly c and $-c$.
- The preimage $B^{-1}([-c, c])$ of the segment $[-c, c]$ is the tree T, the points of $B^{-1}(c)$ being the black vertices and those of $B^{-1}(-c)$ the white ones.

The valency of a vertex is then the multiplicity of B at that point. Such a product exists, and it is unique up to replacing B by $B \circ \psi$, where ψ is an automorphism of $\mathbb{D}$ [1, Theorems 3.4 and 5.2]. Chiu and Ng denote it by B_τ. Since it is only defined up to ψ, its coefficients depend on the choice of a representative. Note also that exchanging the colours of T exchanges c and $-c$, so it does not matter which critical value goes with which colour.

The conjecture. When T is a chain, Chiu and Ng compute the coefficients of B_τ with theta functions [1, Theorem 9.2]. For a general tree, they conjecture [1, Conjecture 11.1]

that there are functions $a_0, \dots, a_n, b_0, \dots, b_m$, meromorphic on $\mathbb{H}$ and algebraic over the field $\mathbb{Q}(\sqrt{k}, \sqrt{k \circ s_n}, (\omega_1 \circ s_n)/\omega_1)$, such that

$$B_\tau(z) = \frac{a_n(\tau)z^n + \dots + a_0(\tau)}{b_m(\tau)z^m + \dots + b_0(\tau)}$$

for every τ on the positive imaginary axis. Since B_τ is only defined up to an automorphism of the disc, this has to be understood for a suitable choice of the representative B_τ.

The result. We say that a function a on $\mathbb{H}$ is *algebraic over* $\mathbb{Q}(c)$ if there is a polynomial $F \in \mathbb{Q}[X, Y]$, of positive degree in Y, such that $F(c(\tau), a(\tau)) = 0$ for all $\tau \in \mathbb{H}$.

Theorem. *There are polynomials P_τ and Q_τ of degree n, with $Q_\tau(0) = 1$, whose coefficients are holomorphic functions of $\tau \in \mathbb{H}$ and are algebraic over $\mathbb{Q}(c)$, with the following property. For every τ on the positive imaginary axis, the rational function $R_\tau = P_\tau/Q_\tau$ is a representative of B_τ, for which 0 is a black vertex of valency m_1 and $R_\tau(1) = 1$. In particular the conjecture of Chiu and Ng is true.*

We came to this question from Blaschke products associated with an ellipse, in the sense of Poncelet. By Theorem Z of [3], a Blaschke product of degree $n \geq 3$ is associated with an ellipse if and only if it has exactly one critical value in the disc, or exactly two critical values in the disc and only simple critical points there. In the second case every vertex of its tree has valency at most 2, so the tree is a chain. This is the case solved by Chiu and Ng, and the theorem deals with all the other trees.

2 The picture

Leaving the disc. A Blaschke product B is a rational function, and it satisfies $B(1/\bar{z}) = 1/\overline{B(z)}$. In other words, what B does outside the disc is the mirror image of what it does inside. On the Riemann sphere, B_τ therefore has four critical values: c and $-c$ in the disc, and their mirror images $1/c$ and $-1/c$ outside. (The mirror image of c is $1/\bar{c}$, which is equal to $1/c$ because c is real.) Over c and over $1/c$ its preimages have the multiplicities $m_1, \dots, m_r$, and over $-c$ and over $-1/c$ they have the multiplicities $\mu_1, \dots, \mu_s$.

If we write $1/c$ instead of $1/\bar{c}$, this description no longer involves complex conjugation, and it still makes sense when c is a complex number. The four points c, $-c$, $1/c$, $-1/c$ are distinct exactly when c belongs to

$$X_0 = \mathbb{C} \setminus \{0,\ 1,\ -1,\ i,\ -i\}.$$

Indeed $c = -c$ means $c = 0$, $c = 1/c$ means $c = \pm 1$, and $c = -1/c$ means $c = \pm i$, and the other coincidences give the same values. We call 0, 1, -1, i and $-i$ the *forbidden values*.

The family $\mathcal{F}(c)$. For $c \in X_0$, let $\mathcal{F}(c)$ be the set of rational functions R of degree n such that:

(i) the preimages of c, and also those of $1/c$, have multiplicities $m_1, \dots, m_r$, and the preimages of $-c$, and also those of $-1/c$, have multiplicities $\mu_1, \dots, \mu_s$,

(ii) $R(0) = c$ with multiplicity m_1, $R(\infty) = 1/c$ with multiplicity m_1, and $R(1) = 1$.

A point of multiplicity m counts for $m-1$ critical points. So condition (i) accounts for $2\sum(m_i - 1) + 2\sum(\mu_j - 1) = 2(n - r) + 2(n - s) = 2n - 2$ critical points, which is the total number of critical points of a rational function of degree n. As a consequence, *the only critical values of*

an element of $\mathcal{F}(c)$ are c, $-c$, $1/c$ and $-1/c$. Condition (ii) is a normalisation. If R satisfies (i), then so does $R \circ h$ for every Möbius map h, and since a Möbius map is determined by the images of three points, condition (ii) removes this freedom.

Lemma 1. *Let τ be on the positive imaginary axis. Then B_τ has a representative in $\mathcal{F}(c(\tau))$. Moreover, every element of $\mathcal{F}(c(\tau))$ which is a Blaschke product with tree T is a representative of B_τ.*

Proof. Let B be a representative of B_τ, and let $x \in \mathbb{D}$ be a black vertex of valency m_1. After composing B with an automorphism of $\mathbb{D}$ which sends 0 to x, we may assume that $x = 0$. The equation $B(z) = 1$ has n solutions, all on the unit circle, and after composing with a rotation we may assume that $B(1) = 1$. Now B satisfies (i), as explained above. It also satisfies $B(0) = c$ with multiplicity m_1, and $B(\infty) = 1/\overline{B(0)} = 1/c$ with the same multiplicity. So B belongs to $\mathcal{F}(c)$. The second assertion follows from the uniqueness of B_τ. □

The covering. We write an element of $\mathcal{F}(c)$ as $R = P/Q$ with $Q(0) = 1$. This is possible, and in only one way, because $R(0) = c$ is finite. Since $R(\infty) = 1/c$ is neither 0 nor ∞, both P and Q have degree n. We identify R with the list of the coefficients of P and Q, which is a point of $\mathbb{C}^{2n+2}$, and we put all the sets $\mathcal{F}(c)$ together:

$$Y = \{(c, R) \ : \ c \in X_0, \ R \in \mathcal{F}(c)\} \subset X_0 \times \mathbb{C}^{2n+2}, \qquad \pi : Y \to X_0, \quad \pi(c, R) = c.$$

The fibre $\pi^{-1}(c)$ is the set $\mathcal{F}(c)$. Figure 1 shows Y above X_0, together with the half-plane $\mathbb{H}$, which is sent into X_0 by $c(\tau)$.

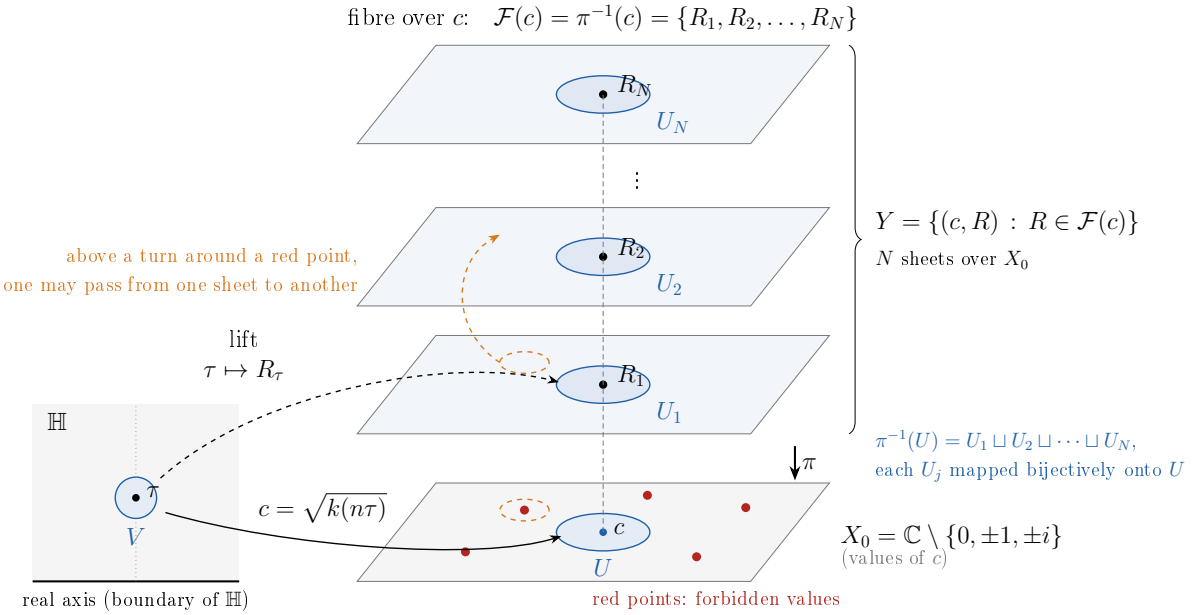


Figure 1: The covering $\pi : Y \to X_0$ and the map $c : \mathbb{H} \to X_0$. Over a small disc U around c lie N disjoint copies $U_1, \ldots, U_N$, one around each element R_j of the fibre $\mathcal{F}(c)$. Turning around a forbidden value can exchange the sheets. The theorem says that $\tau \mapsto R_\tau$ (dashed) is defined on all of $\mathbb{H}$.

The goal. By Lemma 1, when τ is on the imaginary axis, B_τ is a point of Y above $c(\tau)$. We want to extend $\tau \mapsto B_\tau$ to a map $\tau \mapsto R_\tau$ from $\mathbb{H}$ to Y, with $R_\tau \in \mathcal{F}(c(\tau))$, defined without ambiguity at every point of $\mathbb{H}$. This is the dashed arrow of Figure 1. Two facts make it possible.

The sheets of Y can be followed when c moves in X_0 (Proposition 3), and $c(\tau)$ never takes a forbidden value (Lemma 2).

There is a subtle point here. Turning around a forbidden value *does* exchange the sheets of Y. In the example of Section 5, when c turns once around 0, the Blaschke solution is sent to another element of $\mathcal{F}(c)$. As functions of c, the coefficients are therefore multivalued, like $\sqrt{z}$ around 0, and there is nothing we can do about it. But

as functions of τ, through $c(\tau)$, the coefficients are well defined, because no closed loop of $\mathbb{H}$ gives a turn around a forbidden value.

More precisely, if γ is a closed loop in $\mathbb{H}$, then $c\circ\gamma$ is a closed loop in X_0 which can be shrunk to a point without leaving X_0 (see Step 1 below). A turn around a forbidden value can only come from a path of $\mathbb{H}$ which is *not* closed, such as the path from τ_0 to τ_0+1 in Section 5. Arriving on another element of $\mathcal{F}(c)$ at another point of $\mathbb{H}$ is not a contradiction. The exponential function behaves in the same way: $\log w$ is multivalued as a function of w, but $\log(e^z) = z$ is a perfectly good function of z.

3 The proof

Lemma 2. *The function c is holomorphic on $\mathbb{H}$, and $c(\tau) \in X_0$ for every $\tau \in \mathbb{H}$. If τ is on the positive imaginary axis, then $c(\tau)$ is real and $0 < c(\tau) < 1$.*

Proof. By the Jacobi triple product,

$$\vartheta_3 = \prod_{m\geq 1}(1-q^{2m})(1+q^{2m-1})^2, \qquad \vartheta_0 = \prod_{m\geq 1}(1-q^{2m})(1-q^{2m-1})^2,$$
$$\vartheta_2 = 2q^{1/4}\prod_{m\geq 1}(1-q^{2m})(1+q^{2m})^2,$$

where $q^{1/4} = e^{\pi i\tau/2}$. For $\tau \in \mathbb{H}$ we have $0 < |q| < 1$, so the products converge and none of their factors vanishes. The three functions therefore have no zero in $\mathbb{H}$, and neither do the functions $\tau \mapsto \vartheta_j(n\tau)$. It follows that c is holomorphic on $\mathbb{H}$ and never vanishes. Jacobi's identity $\vartheta_3^4 = \vartheta_2^4 + \vartheta_0^4$ [1, (7)] gives

$$1 - c(\tau)^4 = \Big(\frac{\vartheta_0(n\tau)}{\vartheta_3(n\tau)}\Big)^4 \neq 0,$$

so $c^4 \neq 1$, which means that c is not 1, -1, i or $-i$. When τ is on the positive imaginary axis, q is a real number between 0 and 1. The series then show that $\vartheta_2(n\tau)$ and $\vartheta_3(n\tau)$ are positive and that $\vartheta_0(n\tau)$ is real, so $c > 0$ and $1 - c^4 > 0$. □

By a *path* we mean a continuous map defined on $[0,1]$. Let γ be a path in X_0, and let $R_0 \in \mathcal{F}(\gamma(0))$. *Following R_0 along γ* means choosing $R_t \in \mathcal{F}(\gamma(t))$ for $0 \leq t \leq 1$, depending continuously on t and starting from the given R_0. In other words, it means lifting γ to a path $t \mapsto (\gamma(t), R_t)$ in Y.

Proposition 3. *There is an integer N with the following property. Every $c^* \in X_0$ is the centre of a disc $U \subset X_0$ on which there are N holomorphic maps $R_1, \ldots, R_N$ from U to $\mathbb{C}^{2n+2}$, taking distinct values at every point of U, such that*

$$\mathcal{F}(c) = \{R_1(c), \ldots, R_N(c)\} \qquad \text{for every } c \in U.$$

As a consequence, $\pi : Y \to X_0$ is a covering with N sheets, and:

(a) *every $R_0 \in \mathcal{F}(\gamma(0))$ can be followed along every path γ in X_0, in exactly one way,*

(b) *if two paths γ and γ' in X_0 have the same endpoints, and if one can be deformed into the other inside X_0 with the endpoints fixed, then following R_0 along γ and along γ' leads to the same element of $\mathcal{F}(\gamma(1))$.*

The first assertion is proved in Section 4. It says that above U, the set Y is made of the N disjoint graphs $U_j = \{(c, R_j(c)) : c \in U\}$ of Figure 1, each of which is sent bijectively onto U by π. This is the definition of a covering. Properties (a) and (b) are then the path lifting and homotopy lifting properties of coverings (see for instance [1, Lemmas 2.2 and 2.4]). We now prove the theorem.

Step 1: extending to $\mathbb{H}$. We fix τ_0 on the positive imaginary axis, and we choose a representative $R_{\tau_0} \in \mathcal{F}(c(\tau_0))$ of B_{τ_0} (Lemma 1). Let $\tau \in \mathbb{H}$, and let γ be a path in $\mathbb{H}$ from τ_0 to τ. By Lemma 2, $c \circ \gamma$ is a path in X_0, so by Proposition 3(a) we can follow R_{τ_0} along it. We call R_τ the element of $\mathcal{F}(c(\tau))$ that we reach at the end.

R_τ does not depend on the path. Let γ' be another path in $\mathbb{H}$ from τ_0 to τ. Since $\mathbb{H}$ is convex, the paths $\gamma_u = (1-u)\gamma + u\gamma'$, for $0 \le u \le 1$, deform γ into γ' inside $\mathbb{H}$ without moving the endpoints. Then the paths $c \circ \gamma_u$ deform $c \circ \gamma$ into $c \circ \gamma'$ without moving the endpoints, and by Lemma 2 they stay inside X_0 all the time. In other words, they never pass through a forbidden value. By Proposition 3(b), following R_{τ_0} along $c \circ \gamma$ or along $c \circ \gamma'$ leads to the same element. This is exactly what the sentence in the box means: the closed loop made of γ followed by γ' backwards has an image which can be shrunk to a point inside X_0.

The coefficients of R_τ are holomorphic. Let $\tau_1 \in \mathbb{H}$. Take a disc U around $c(\tau_1)$ as in Proposition 3, and a disc $V \subset \mathbb{H}$ around τ_1 such that $c(V) \subset U$. Let j be the index for which $R_{\tau_1} = R_j(c(\tau_1))$. For $\tau \in V$, we can reach τ by a path from τ_0 to τ_1 followed by the segment from τ_1 to τ. Along this segment, $R_j \circ c$ follows R_{τ_1}, so by uniqueness it is the way we follow it. Hence $R_\tau = R_j(c(\tau))$ on V, and this depends holomorphically on τ.

Step 2: on the imaginary axis, R_τ is a representative of B_τ. For a rational function R, we call $R^*(z) = 1/\overline{R(1/\bar{z})}$ its mirror image. It is a rational function of the same degree, whose coefficients are obtained from those of R by taking complex conjugates and reversing their order. If R is a Blaschke product, then $R^* = R$.

If c is real and $R \in \mathcal{F}(c)$, then $R^* \in \mathcal{F}(c)$. Indeed, $R^*(z) = w$ is equivalent to $R(1/\bar{z}) = 1/\bar{w}$, and the map $z \mapsto 1/\bar{z}$ preserves multiplicities. As c is real, the preimages of c under R^* are the mirror images of the preimages of $1/c$ under R, and they have the same multiplicities $m_1, \dots, m_r$. The same argument applies to $1/c$, $-c$ and $-1/c$, which gives condition (i). For condition (ii), we have $R^*(0) = 1/\overline{R(\infty)} = c$ and $R^*(\infty) = 1/\overline{R(0)} = 1/c$, with multiplicity m_1, and $R^*(1) = 1/\overline{R(1)} = 1$.

On the axis, $R_\tau^* = R_\tau$. Let γ be the segment of the imaginary axis from τ_0 to τ. Along this segment c is real (Lemma 2). So $t \mapsto R_{\gamma(t)}$ and $t \mapsto R^*_{\gamma(t)}$ are two ways of following an element of $\mathcal{F}(c(\tau_0))$ along $c \circ \gamma$, and they start from the same element, because R_{τ_0} is a Blaschke product. By the uniqueness in Proposition 3(a), they are equal. Hence $R_\tau(z) = 1/\overline{R_\tau(1/\bar{z})}$, and for $|z| = 1$ this says that $|R_\tau(z)| = 1$.

R_τ is a Blaschke product. The n zeros of R_τ are the roots of P_τ, so they move continuously with τ. None of them can be on the unit circle, where $|R_\tau| = 1$. At τ_0 they are all in $\mathbb{D}$, so they stay in $\mathbb{D}$ for every τ on the axis. Let B be the Blaschke product with the same zeros. The poles of R_τ are the mirror images of its zeros, and so are those of B. So R_τ/B is a rational function with no zero and no pole, that is, a constant. This constant has modulus 1, since $|R_\tau| = |B| = 1$ on the circle.

The tree of R_τ is T. The critical values of R_τ in $\mathbb{D}$ are c and $-c$, since $1/c$ and $-1/c$ lie outside the disc. The vertices of the tree $R_\tau^{-1}([-c,c])$ are the preimages of c and $-c$. They move continuously with τ and keep their multiplicities, so no two of them can meet. The edges are the n lifts of the open segment, and they also move continuously. As a result, neither the way the edges are attached to the vertices nor their cyclic order around each vertex can change along the axis. The tree of R_τ is therefore the tree of R_{τ_0}, which is T.

By Lemma 1, R_τ is a representative of B_τ, and by condition (ii) it is the one described in the theorem.

Step 3: algebraicity. Let $a(\tau)$ be one of the coefficients of P_τ or Q_τ. By Section 4, for $c \in X_0$ the elements of $\mathcal{F}(c)$ are given by the genuine solutions (u,c) of a system of polynomial equations with integer coefficients, and the condition of being genuine is expressed by polynomial inequations with integer coefficients. The coefficient a is one of the coordinates of u. Let $W \subset \mathbb{C}^2$ be the set of all the pairs (c,a) obtained from genuine solutions. By elimination theory, W is again described by polynomial equations and inequations with rational coefficients. Since it has finitely many points above each c, it is not dense in $\mathbb{C}^2$, so there is a nonzero polynomial $G \in \mathbb{Q}[X,Y]$ which vanishes on W. As $(c(\tau), a(\tau))$ belongs to W, we get $G(c(\tau), a(\tau)) = 0$ for all $\tau \in \mathbb{H}$.

Write $G(X,Y) = g(X)F(X,Y)$, where $g \in \mathbb{Q}[X]$ and F has no factor depending on X alone. The function c is holomorphic and not constant, so $g(c(\tau))$ vanishes only at isolated points. Hence $F(c(\tau), a(\tau)) = 0$ on a dense subset of $\mathbb{H}$, and therefore everywhere, by continuity. Finally, F is not constant (otherwise $g(c(\tau))$ would be zero for all τ), and it has no factor depending on X alone, so it has positive degree in Y.

Conclusion. The coefficients of P_τ and Q_τ are holomorphic on $\mathbb{H}$ (Step 1) and algebraic over $\mathbb{Q}(c)$ (Step 3), and $R_\tau = P_\tau/Q_\tau$ is a representative of B_τ on the imaginary axis (Step 2). Since $c = \sqrt{k \circ s_n}$ belongs to the field $\mathbb{Q}(\sqrt{k}, \sqrt{k \circ s_n}, (\omega_1 \circ s_n)/\omega_1)$, these coefficients are meromorphic functions on $\mathbb{H}$ which are algebraic over this field. This is Conjecture 11.1 of [1], with $m = n$. □

4 Proof of Proposition 3

It remains to prove the first assertion of Proposition 3.

The equations. Let $c \in X_0$ and let $R = P/Q$ be an element of $\mathcal{F}(c)$, with $Q(0) = 1$. A *ramification point* of R is a point of multiplicity at least 2. By (i), it lies above one of the four values c, $-c$, $1/c$ and $-1/c$. A point $x \in \mathbb{C}$ where R takes the value w with multiplicity m is a root of order m of the polynomial $P - wQ$. We call *free* the ramification points other than 0 and ∞. We denote them by $x_1, \dots, x_f$, with multiplicities $e_1, \dots, e_f$, above the values $w_1, \dots, w_f$. The list of the pairs (e_k, w_k) is prescribed by (i) and (ii), so it is the same for all the elements of $\mathcal{F}(c)$. The function R then satisfies

$$\begin{aligned}
&(P - w_k Q)^{(j)}(x_k) = 0 && (1 \le k \le f,\ 0 \le j \le e_k - 1),\\
&(P - cQ)^{(j)}(0) = 0 && (0 \le j \le m_1 - 1),\\
&cP - Q \text{ has degree at most } n - m_1,\\
&P(1) = Q(1) \quad \text{and} \quad Q(0) = 1.
\end{aligned} \tag{1}$$

The second line says that $R(0) = c$ with multiplicity at least m_1, and the third line, which says that the coefficients of $z^n, \dots, z^{n-m_1+1}$ in $cP - Q$ are zero, means that $R(\infty) = 1/c$ with

multiplicity at least m_1. Let u be the list made of the $2n+2$ coefficients of P and Q and of the points $x_1, \dots, x_f$. Then (1) is a system $E(u,c) = 0$ of polynomial equations in u and c with integer coefficients (we multiply by c the equations in which $w_k = \pm 1/c$). We say that a solution is *genuine* if $c \in X_0$, if P and Q have degree n and no common root, and if the points $0, x_1, \dots, x_f$ are all distinct.

Genuine solutions and elements of $\mathcal{F}(c)$. An element of $\mathcal{F}(c)$ whose free ramification points have been numbered gives a genuine solution. Conversely, let u be a genuine solution. Then $R = P/Q$ has degree n, and it has at least the prescribed multiplicities at the distinct points $0, \infty, x_1, \dots, x_f$. By the count of Section 2, these multiplicities already account for $2n-2$ critical points. So they are exact, R has no other critical point, and R satisfies (i) and (ii). The elements of $\mathcal{F}(c)$ are therefore the functions P/Q given by the genuine solutions, and each element is given by finitely many of them, one for each numbering of its free ramification points.

There are as many equations as unknowns. There are $2n+2+f$ unknowns. The number of equations is $\sum_k e_k + 2m_1 + 2$. By the count of Section 2, $\sum_k (e_k - 1) + 2(m_1 - 1) = 2n - 2$, so $\sum_k e_k = 2n - 2m_1 + f$, and the number of equations is also $2n+2+f$.

(a) Locally, the sheets are copies of X_0. We use the implicit function theorem, in the following form. *Let $E(u,c) = 0$ be a system of k holomorphic equations in k unknowns u, depending on a parameter c. If it has a solution (u^*, c^*) at which the matrix $J = \partial E/\partial u$ is invertible, then there are a disc U around c^* and a neighbourhood V of u^* such that, for every $c \in U$, the system has exactly one solution $u(c)$ in V. Moreover $u(c)$ depends holomorphically on c.*

Our system is polynomial, and it has as many equations as unknowns. It remains to show that J is invertible at every genuine solution (u^*, c^*), or in other words that the linear system $J\,\delta u = 0$ has no solution other than $\delta u = 0$. Let $\delta u = (\delta P, \delta Q, \delta x_1, \dots, \delta x_f)$ be a solution, where δP and δQ are polynomials of degree at most n. Since $J\,\delta u$ is the derivative of $E(u^* + \varepsilon\,\delta u, c^*)$ at $\varepsilon = 0$, the equations $J\,\delta u = 0$ read as follows, with $S_k = P - w_k Q$ and $\delta S_k = \delta P - w_k\,\delta Q$:

$$\begin{aligned}
&\delta S_k^{(j)}(x_k) + S_k^{(j+1)}(x_k)\,\delta x_k = 0 && (1 \le k \le f,\ 0 \le j \le e_k - 1),\\
&(\delta P - c\,\delta Q)^{(j)}(0) = 0 && (0 \le j \le m_1 - 1),\\
&c\,\delta P - \delta Q \text{ has degree at most } n - m_1,\\
&\delta P(1) = \delta Q(1) \quad \text{and} \quad \delta Q(0) = 0.
\end{aligned} \tag{2}$$

Intuitively, δu is a way of moving R, to first order, without moving c. We show that this is impossible. The idea is that such a motion can only be a change of variable, and the normalisation does not allow changes of variable. We put

$$\delta R = \frac{\delta P\, Q - P\,\delta Q}{Q^2} \qquad \text{and} \qquad v = \frac{\delta R}{R'}.$$

Since $R(z + \varepsilon v(z)) = R(z) + \varepsilon\,\delta R(z) + O(\varepsilon^2)$, the variation δR is the effect of the change of variable $z \mapsto z + \varepsilon v(z)$.

The function v has no pole in $\mathbb{C}$. We have $R - w_k = S_k/Q$, so $R' = (S_k' Q - S_k Q')/Q^2$ and $\delta R = (\delta S_k\, Q - S_k\,\delta Q)/Q^2$. At x_k, the polynomial S_k vanishes to order exactly e_k, so R' vanishes to order exactly $e_k - 1$. Since $S_k^{(j+1)}(x_k) = 0$ for $j \le e_k - 2$, the first line of (2) says that δS_k vanishes at x_k to order at least $e_k - 1$, and so does δR. So v is holomorphic at x_k. Comparing the leading terms, and using the equation for $j = e_k - 1$, we also get

$$v(x_k) = \frac{\delta S_k^{(e_k-1)}(x_k)}{S_k^{(e_k)}(x_k)} = -\delta x_k.$$

At 0, the second line of (2) says that $\delta P - c\,\delta Q$ vanishes to order at least m_1. So δR vanishes to order at least m_1, while R' vanishes to order exactly $m_1 - 1$. Hence v is holomorphic at 0 and $v(0) = 0$. At a root of Q, the function R has a simple pole, because ∞ is not a critical value. There R' has a pole of order 2 and δR a pole of order at most 2, so v is holomorphic. At every other point of $\mathbb{C}$, R' does not vanish.

The function v is a polynomial of degree at most 1. We use the coordinate $\zeta = 1/z$ near ∞. In this coordinate, $R - 1/c$ vanishes at $\zeta = 0$ to order exactly m_1, so $dR/d\zeta$ vanishes to order exactly $m_1 - 1$. On the other hand,

$$\delta R = \frac{c\,\delta P - \delta Q}{cQ} - \frac{(cP - Q)\,\delta Q}{cQ^2}$$

vanishes to order at least m_1, by the third lines of (1) and (2). So $\delta R/(dR/d\zeta) = -\zeta^2 v(1/\zeta)$ is holomorphic at $\zeta = 0$ and vanishes there. Now, if v is holomorphic on $\mathbb{C}$ and $\zeta^2 v(1/\zeta)$ is holomorphic at 0, then v is a polynomial of degree at most 2, say $v = v_0 + v_1 z + v_2 z^2$. Since $\zeta^2 v(1/\zeta) = v_0\zeta^2 + v_1\zeta + v_2$ vanishes at $\zeta = 0$, we get $v_2 = 0$.

The function v is zero. Since $v(0) = 0$, we have $v_0 = 0$. The last line of (2) gives $\delta R(1) = (\delta P(1) - \delta Q(1))/Q(1) = 0$, where $Q(1) = P(1) \neq 0$. Moreover $R'(1) \neq 0$, because 1 is not a critical value (recall that $c \neq \pm 1$). So $v(1) = 0$, which gives $v_1 = 0$.

Conclusion: $\delta u = 0$. From $v = 0$ we get $\delta R = 0$, that is, $\delta P\, Q = P\, \delta Q$. As P and Q have no common root, Q divides δQ. Since δQ has degree at most n, which is the degree of Q, there is a constant κ such that $\delta Q = \kappa Q$, and then $\delta P = \kappa P$. The conditions $\delta Q(0) = 0$ and $Q(0) = 1$ give $\kappa = 0$. Finally $\delta x_k = -v(x_k) = 0$.

So the implicit function theorem applies at every genuine solution (u^*, c^*). For c close to c^*, the solution $u(c)$ is still genuine, because the conditions defining genuine solutions are open conditions.

(b) No sheet is lost. The implicit function theorem says nothing about what happens far from a given solution. A priori, when c tends to a point of X_0, a solution could go to infinity or stop being genuine. We rule this out by counting.

The number of elements of $\mathcal{F}(c)$ is finite and does not depend on $c \in X_0$. Let S be a set of four points of the sphere. By Riemann's existence theorem [2], the rational functions of degree n whose critical values lie in S, up to Möbius changes of variable $R \mapsto R \circ h$, correspond to the transitive homomorphisms from the fundamental group of the sphere minus S to the symmetric group S_n, up to conjugation. Under this correspondence, the multiplicities of the preimages of a point of S are the lengths of the cycles of the permutation associated with a small loop around this point. So, with prescribed multiplicities, there are finitely many classes. Their number does not depend on S, because a homeomorphism of the sphere carrying the points of S to those of another set S', in a given order, carries small loops to small loops.

Now fix a class and a function R in it. The elements of $\mathcal{F}(c)$ in this class are the functions $R \circ h$, where h is the Möbius map sending 0, ∞ and 1 to a point of multiplicity m_1 above c, a point of multiplicity m_1 above $1/c$, and one of the n points above 1. (There are exactly n points above 1, because 1 is not a critical value.) These three points are distinct, since c, $1/c$ and 1 are distinct, so each choice gives exactly one map h. The number of choices depends only on T. Two choices give the same function $R \circ h$ exactly when they differ by a Möbius map g such that $R \circ g = R$, and the number of such maps g is determined by the monodromy of the class. This proves the claim. We call N this number.

End of the proof. Let $c^* \in X_0$, and let $R_1^*, \ldots, R_N^*$ be the elements of $\mathcal{F}(c^*)$. For each j, we choose a genuine solution giving R_j^*. By (a), it extends to a holomorphic family of

genuine solutions over a disc around c^*, which gives a holomorphic map $c \mapsto R_j(c) \in \mathcal{F}(c)$ with $R_j(c^*) = R_j^*$. Since the R_j^* are distinct, there is a disc $U \subset X_0$ around c^* on which all the maps R_j are defined and take distinct values. For $c \in U$, the set $\mathcal{F}(c)$ contains the N distinct elements $R_1(c), \dots, R_N(c)$, and it has exactly N elements. Therefore $\mathcal{F}(c) = \{R_1(c), \dots, R_N(c)\}$. □

5 An example: the fork

Take $n = 4$, and the tree with black vertices of valencies 3 and 1 and white vertices of valencies 2, 1 and 1. It is the smallest tree which is neither a chain nor a star. Here $c = \sqrt{k(4\tau)}$. For a complex number t, we put

$$G_t(z) = z^3 \, \frac{z - t}{1 - tz} \qquad \text{and} \qquad R = \frac{G_t + c}{1 + c\, G_t}.$$

The function R takes the value c at 0, with multiplicity 3, and at t. It takes the value $1/c$ at ∞, with multiplicity 3, and at $1/t$. It also satisfies $R(1) = 1$ and $R(1/z) = 1/R(z)$. Besides 0 and ∞, R has two simple critical points, exchanged by $z \mapsto 1/z$, so their critical values are inverse to each other. Hence R belongs to $\mathcal{F}(c)$ if and only if one of these two critical values is $-c$, which means that $(G_t + c) + c(1 + c\, G_t)$ has a double root. Computing the discriminant of its numerator, and removing the factor $-4c^2(c^2+1)^2$, which does not vanish on X_0, we find the condition

$$\Phi(t, c) = 8c(c^2+1)\, t^6 + (27c^4 + 12c^3 + 162c^2 + 12c + 27)\, t^4 + 384c(c^2+1)\, t^2 - 512c(c^2+1) = 0.$$

The functions obtained in this way are only some of the elements of $\mathcal{F}(c)$: those for which the simple point above $1/c$ is the inverse of the simple point above c. For τ on the imaginary axis, B_τ has a representative of this form with $0 < t < 1$, obtained by rotating the disc so that the simple black vertex lies on the positive real axis. When we follow this representative as in Step 1, we stay among these functions. Indeed, the product of the two simple points is a holomorphic function of τ which equals 1 on the axis, so it equals 1 everywhere.

Proposition 3 on the example. The discriminant of Φ with respect to t is, up to a constant factor, $c^4(c-1)^8(c+1)^{16}(c^2+1)^4$, and its leading coefficient is $8c(c^2+1)$. So for $c \in X_0$ the equation has six distinct finite roots, and near every point of X_0 they are holomorphic functions of c. Two roots can only meet, and a root can only go to infinity, at the forbidden values.

The sentence in the box, on the example. For $\tau_0 = 0.2i$ we find $c_0 = c(\tau_0) = 0.561870$, and the six roots are ± 0.998965, $\pm 2.643116\, i$ and $\pm 3.029867\, i$. The root $t = 0.998965$ gives a representative of B_{τ_0}. Figure 2 shows what happens when we follow it numerically along two paths starting at τ_0. Along a closed loop in $\mathbb{H}$ (in blue), c describes a loop which does not turn around any forbidden value, and t comes back to its initial value. Along the path from τ_0 to $\tau_0 + 1$ (in orange), which is not closed, c turns once around 0 and comes back to c_0, while t ends on another root, $2.643116\, i$. There is no contradiction here: τ_0 and $\tau_0 + 1$ are two different points of $\mathbb{H}$, and the function $t(\tau)$ simply takes different values at these two points.

References

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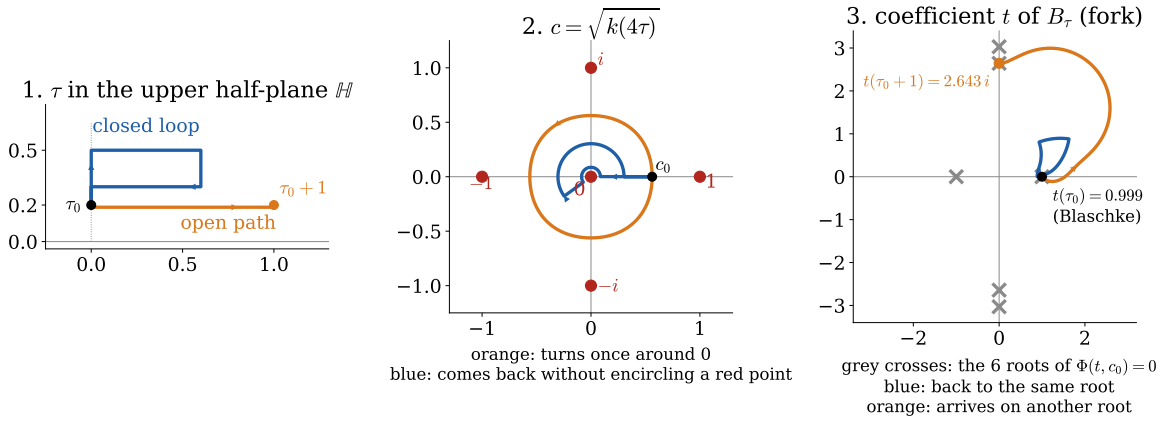


Figure 2: The fork. Left: a closed loop (blue) and an open path (orange) in $\mathbb{H}$, both starting at $\tau_0 = 0.2i$. Middle: their images in the plane of c, where the red points are the forbidden values. Right: the root t followed along both paths. The grey crosses are the six roots of $\Phi(t, c_0) = 0$.

[3] M. Baptiste, *Blaschke products associated to ellipses*, https://cursedmaths.github.io/.